Maximum Level Sum of a Binary Tree (medium)

Problem Statement

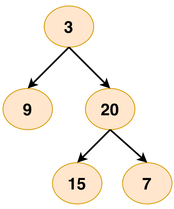
Given a `root` of the binary tree, return the s`mallest` level `x`, such that sum of all node's values at level `x` has maximum sum.

The level of a binary tree starts at 1 for the root node, and increases by 1 for each subsequent depth.

Examples

Example 1:

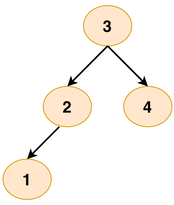
- **Input**: [3,9,20,null,null,15,7]



- **Expected Output**: 2
- **Justification**: The sum of each level is: Level 1: 3, Level 2: 29, Level 3: 22. The second level has the maximum sum, so the output is **2**.

Example 2:

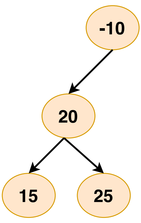
- **Input**: [1,2,3,4,5,null,7]



- **Expected Output**: 3
- **Justification**: The sums are Level 1: 1, Level 2: 5, Level 3: 16. The third level has the highest sum, thus the output is **3**.

Example 3:

- **Input**: [-10,20,null,15,25]



- **Expected Output**: 3
- **Justification**: The sums are Level 1: -10, Level 2: 20, Level 3: 40. The third level's sum is the highest, making the output **3**.

Try it yourself

Try solving this question here:

Java

```java
import java.util.LinkedList;
import java.util.Queue;

// class TreeNode {
//     int val;
//     TreeNode left;
//     TreeNode right;
//     TreeNode(int x) { val = x; } // Constructor to create a new node
// }

public class Solution {
    public int maxLevelSum(TreeNode root) {
        // ToDo: Write Your Code Here.
        return 0; // Return the level with maximum sum
    }
}
```

Test Case | Run Code Result | Stdout

[3,9,20,null,null,15,7]

Use Example Testcases | Run | Submit